Homework Assignment

Update Employee Data

Unit 6 - Lesson 2

💼 Your Mission: Year-End Bonuses!

The company is giving everyone a $2000 year-end bonus! Update the employee database to reflect the new salaries.

Your Tasks:

1. Load all employee data from `employees.csv`
2. Add $2000 to each employee's salary
3. Save the updated data back to `employees.csv`
4. Print confirmation messages showing progress

🎯 Skills You'll Use

📖 **Reading CSV** — Load existing employee data

✏️ **Modifying Data** — Update salary values in a loop

💾 **Writing CSV** — Save changes back to the file

This is the Read → Modify → Write pattern!

The employees.csv File

Your file contains these columns:

```
employee_id,name,department,salary,hire_date
E001,Jennifer Martinez,Sales,65000,2019-03-15
E002,Michael Chen,Marketing,70000,2018-07-22
E003,Sarah Johnson,IT,78000,2020-01-10
...
```

Focus on updating the "salary" column for each employee

📋 Step-by-Step Approach

Step 1: Import csv module

Step 2: Open employees.csv in READ mode, load all data with DictReader, convert to list

Step 3: Print how many employees were loaded

Step 4: Loop through each employee, add 2000 to their salary (remember to convert to int first!)

Step 5: Print confirmation that bonuses were added

📋 Step-by-Step Approach (continued)

Step 6: Open employees.csv in WRITE mode

Step 7: Create DictWriter with fieldnames: employee_id, name, department, salary, hire_date

Step 8: Write the header

Step 9: Write all the updated employee records

Step 10: Print confirmation that data was saved

💻 Expected Console Output

When you run your program, you should see:

```
Loaded 25 employees
Added $2000 bonus to each employee
Saved updated data!
```

Simple, clear, and professional output! ✨

💡 Hints & Tips

💡 **Hint 1: Salary values are strings in the CSV. Convert with int() before adding!**

💡 **Hint 2: Use list() to convert the DictReader to a list you can modify**

💡 **Hint 3: The assignment operator works fine: employee['salary'] = int(employee['salary']) + 2000**

💡 **Hint 4: Don't forget newline="" when opening files for writing!**

⚠️ Common Mistakes to Avoid

- ❌ Forgetting to convert salary to int before adding
- ❌ Opening file in 'a' mode instead of 'w' mode for the final write
- ❌ Forgetting to call writeheader()
- ❌ Not including all fieldnames in the correct order

📤 What to Submit

- ✅ **homework.py** — Your Python code file
- ✅ **employees.csv** — Your updated CSV file (with new salaries)
- ✅ **screenshot.png** — Screenshot of your program output (optional)

Make sure your code is well-commented and easy to read!

📊 Grading Rubric

Criteria	Points
Correctly loads employees.csv	2 pts
Updates all salaries (+$2000)	4 pts
Saves data back correctly	4 pts
Total	**10pts**